

Statistics Project: Confidence Intervals — History, Theory, and Simulation

Background

While point estimation provides a single “best guess” for an unknown parameter, early statisticians quickly recognized its limitations: a point alone says nothing about uncertainty. This led to the development of interval estimation, which quantifies how precise an estimate is and how confident we are that it captures the true parameter. In this project, you will explore the theory, historical foundations, and finite-sample behavior of confidence intervals through derivation and simulation.

Part A: Historical and Conceptual Foundations

Question 1.

Briefly address the following:

- What is a confidence interval?
- Why was interval estimation an important advancement beyond point estimation?
- Describe the historical contributions of Jerzy Neyman and Ronald Fisher to inference and uncertainty quantification.

Question 2.

- What does it mean for a confidence interval to have $(1 - \alpha)100\%$ coverage?
- Carefully explain the frequentist interpretation of a confidence interval.
- Identify and explain one common misconception about confidence intervals.

Part B: Confidence Intervals in a Simple Model

Question 3.

Let $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)$, where p is unknown.

- Derive the Wald confidence interval for p .
- State the Clopper–Pearson (exact) confidence interval and explain how it differs from the Wald interval.
- Briefly discuss why the Wald interval may perform poorly for small n or when p is near 0 or 1.

Part C: Simulation and Coverage Probability

Question 4.

Fix a value of p (for example, $p = 0.3$ or $p = 0.6$).

For each $n \in \{10, 50, 100, 500, 1000\}$:

- Simulate at least 1,000 samples of size n .
- Construct a 95% Wald confidence interval for p from each sample.
- Record whether the true value of p lies within the interval.

Estimate the empirical coverage probability and analyze how coverage changes as n increases.

Part D: Interval Width and Precision

Question 5.

Using the simulations from Part C:

- Estimate the average length of each confidence interval for each n .
- Create plots showing how interval width changes as sample size increases.

Discuss precision and the trade-off between confidence level and interval width.

Part E: A Continuous Distribution Case Study

Question 6.

Let $X_1, \dots, X_n \sim \text{Exponential}(\theta)$, where $\theta > 0$.

- Derive a large-sample confidence interval for θ using the asymptotic distribution of the MLE. (Normal CI)
- Derive a pivotal quantity-based confidence interval
- Compare these intervals conceptually.

Question 7.

Repeat the simulation study for the exponential model.

- Estimate coverage probabilities.
- Compare interval length and finite-sample performance.